

FORMATION OF PLANETARY SYSTEMS: A SIMPLE SYNTHETICAL APPROACH

GIANFRANCO MAGNI

Laboratorio di Astrofisica Spaziale, Frascati, Italy

and

PAOLO PAOLICCHI

Osservatorio Astronomico di Brera, Merate (Co), Italy

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Abstract. It is shown that by means of elementary assumptions it is possible to obtain models of planetary systems not too different from the observed case. Such assumptions are shown to be the bulk of more detailed models as those by Dole (1970) and Isaacman and Sagan (1977).

An accurate fit with the features of the Solar System can be obtained only by *ad hoc* assumptions on various physical quantities and depends also on the possibility of taking into account stochastic processes in the numerical code.

We discuss various cases and attempt to derive best fit values for the free parameters of the model. The hypothesis of a moderate mass depletion in the region of outer planets is strongly supported by our results.

1. Introduction

One of the most striking approaches to the problem of the formation of the Solar System is to derive a general model of planetary accumulation by the *a priori* assumption of a primaeval rotating nebula in which condensation nuclei (planetesimals) are formed.

In particular this approach is analyzed in the papers by Dole (1970) and more recently by Isaacman and Sagan (1977) and Vityazev *et al.* (1978).

In this paper we want to point out that few elementary assumptions can be recognized to be the basis of models such as those of Dole (1970) and Isaacman and Sagan (1977). By means of these assumptions alone we can obtain realistic synthetical planetary systems and fit the Titius-Bode law, its deviations due to the planetary masses (Farinella and Paolicchi, 1977) and the relation of the mean planetary spacing with the mean planetary mass (Colombo, 1975).

We also show that, in order to avoid *ad hoc* hypotheses on the initial physical conditions and on the accumulation process, it is necessary to assume mass depletion in the outer regions. Moreover, eccentricity effects and those connected to stochastic processes (fluctuations, collisions and so on) should probably be taken into account.

2. Basic Assumptions

We start with an axially symmetrical rotating disk of dust and gas, with a surface density $\sigma(r)$, and a ratio $\sigma_{\text{gas}}/\sigma_{\text{dust}} = 50$, near to solar values.

Let us assume that dust initially, and planets after the process of formation, maintain circular orbits. Let us also assume the following.

- (1) The mechanism of planets formation is the same along the system.
- (2) All the dust is captured by the planets.
- (3) Every planet captures matter in a ring defined by

$$x/r = \theta(m/M)^\beta, \quad (1)$$

where x is the ring width, r the distance of the planet from the Sun determined by angular momentum conservation, m and M are the planet and the Sun masses, θ and β are two free parameters (Dole assumes $\theta = 1$, $\beta = \frac{1}{4}$).

- (4) The gas capture can be considered in three ways.

- (a) Planets capture only the dust.
- (b) Planets capture all the dust and the gas.
- (c) Planets capture all the dust and part of the gas.

Cases (a) and (b) can be considered equivalent, formally varying the density by a factor 51.

3. One Component Model

Assuming that the disc mass is much lower than that of the Sun, the velocities are nearly Keplerian; so that we have

$$m_i = 2\pi \int_{R_i}^{R_{i+1}} \sigma(r) r dr, \quad (2)$$

$$r_i = \left[\frac{\int_{R_i}^{R_{i+1}} \sigma(r) r^{3/2} dr}{\int_{R_i}^{R_{i+1}} \sigma(r) r dr} \right]^2, \quad (3)$$

where R_i and R_{i+1} are the boundaries of the ring. If we define

$$I_\sigma(i, j, h) = \int_{R_i}^{R_j} \sigma(r) r^h dr, \quad (4)$$

Equation (1) becomes

$$(R_{i+1} - R_i) \left[\frac{I_\sigma(i, i+1, 1)}{I_\sigma(i, i+1, \frac{3}{2})} \right]^2 = \theta [(2\pi/M) I_\sigma(i, i+1, 1)]^\beta. \quad (5)$$

If $R_k = \alpha_k R$ ($\alpha_k \leq 1$, R is the nebular radius) we have

$$\begin{aligned} I_\sigma(i, i+1, h) &= R^{h+1} \int_{\alpha_i}^{\alpha_{i+1}} \sigma(\alpha) \alpha^h d\alpha \\ &\equiv R^{h+1} \mathcal{J}(i, i+1, h), \end{aligned} \quad (6)$$

where $\mathcal{S}$ has the dimension of a surface density. Finally, we have

$$\left[\frac{\mathcal{S}(i, i+1, 1)}{\mathcal{S}(i, i+1, \frac{3}{2})} \right]^2 (\alpha_{i+1} - \alpha_i) = \theta [(2\pi R^2/M) \mathcal{S}(i, i+1, 1)]^\beta. \quad (7)$$

Equation (7) is a recurrent relation between α_i and α_{i+1} for every given set of θ , β and $\sigma(r)$.

Let us assume that the surface density has a power law dependence on α : $\sigma(\alpha) = \sigma_0 \alpha^{-\chi}$ ($\chi < 2$) (this assumption is in sufficient agreement with the present mass distribution in the solar system (Weidenschilling, 1977)). Then

$$\sigma(\alpha) = \frac{1}{2} \bar{\sigma} (2 - \chi) \alpha^{-\chi}, \quad (8)$$

where $\bar{\sigma} = 2\sigma_0/(2 - \chi)$ is the mean density. We have

$$\mathcal{S}(i, i+1, h) = \bar{\sigma} \frac{2 - \chi}{2(1 - h - \chi)} (\alpha_{i+1}^{1+h-\chi} - \alpha_i^{1+h-\chi}), \quad (9)$$

so that Equation (7) becomes

$$\begin{aligned} & [(2.5 - \chi)/(2 - \chi)]^2 (\alpha_{i+1}^{2-\chi} - \alpha_i^{2-\chi})^{2-\beta} (\alpha_{i+1} - \alpha_i) \\ & \times (\alpha_{i+1}^{2.5-\chi} - \alpha_i^{2.5-\chi})^{-2} = \theta \mu^\beta, \end{aligned} \quad (10)$$

where $\mu = \pi R^2 \bar{\sigma}/M$ is the disk-Sun mass ratio.

We define

$$\alpha_{i+1} = \alpha_i (1 + \delta\omega). \quad (11)$$

If $\delta\omega \ll 1$ (narrow rings), Equation (9) gives

$$\delta\omega = \theta^{1/(1-\beta)} (2 - \chi)^{\beta/(1-\beta)} \alpha_i^{\beta(2-\chi)/(1-\beta)} \mu^{\beta/(1-\beta)}. \quad (12)$$

If α_1 is the innermost boundary of the rings, the number of resulting planets can be approximately given by

$$\begin{aligned} N &= \int_{\alpha_1}^1 \frac{1}{\alpha} \frac{d\alpha}{\delta\omega(\alpha)} = \frac{1-\beta}{\beta} \theta^{1/(\beta-1)} (2 - \chi)^{1/(\beta-1)} \times \\ &\times [\alpha_1^{\beta(2-\chi)/(\beta-1)} - 1] \mu^{\beta/(\beta-1)} = Cf(\alpha_1)g(\mu). \end{aligned} \quad (13)$$

Equation (12) implies that if $\beta(2 - \chi)/(1 - \beta) \ll 1$, $\delta\omega$ is independent on α_i , i.e., $\alpha_{i+1} = A\alpha_i$ (Titius-Bode law); since $A^{N-1}\alpha_1 = 1$, we have

$$A = \alpha_1^{-1/(N-1)}. \quad (14)$$

Using Equation (13) to express N as function of the mean mass of the planets $\gamma = \mu/N$ we have

$$N^{1/(1-\beta)} = Cf(\alpha_1)g(\gamma). \quad (15)$$

Substituting Equation (15) into Equation (14) we can infer that A depends on the mean mass of the system (Colombo, 1975). Finally, since by Equation (2)

$$\begin{aligned}
m_i &= \mu(\alpha_{i+1}^{2-\chi} - \alpha_i^{2-\chi}) \\
&= \mu\alpha_i^{2-\chi}(2 - \chi)\delta\omega \\
&\propto \alpha_i^{(2-\chi)/(1-\beta)} \\
&\propto \delta\omega^{1/\beta} \quad (\text{from Equation (12)}),
\end{aligned} \tag{16}$$

we conclude that: (i) the masses increase with the distance from the Sun, (ii) the deviations from the Titius-Bode law depend on the planetary mass (Farinella and Paolicchi, 1977).

Table I shows the N values given by Equation (13) assuming $\theta = 1$, $\beta = \frac{1}{4}$ (Dole values), $\chi = \frac{3}{2}$ and 1 for $\alpha_1 = 0.01, 0.03$ and 0.1 and $\mu = 10^{-4}, 10^{-3}$ and 10^{-2} .

We remark that these values are upper limitations for the N values which can be obtained by various gas-capture assumptions.

TABLE I
The approximate N values given by Equation (13) for various α_1 and μ values in both cases $\chi = \frac{3}{2}$ (upper value) and $\chi = 1$ (lower value)

	$\alpha_1 = 0.01$	$= 0.03$	$= 0.1$
$\chi = \frac{3}{2}$	188	130	76
$\mu = 10^{-4}$			
$\chi = 1$	235	143	75
	87	60	35
10^{-3}			
	109	67	35
	41	28	16
10^{-2}			
	51	31	16

4. Two Component Model

To obtain realistic models we must take into account the gas capture. A qualitative discussion of this process can be afforded under various assumptions. Dole (1970) assumes that, after a complete dust capture, the gas can be accreted by the i th planet according to the iterative formula

$$\sigma^{(j+1)} = \begin{cases} \sigma^{(j)} & \text{if } m_i^{(j)} < m_0, \\ 51\sigma^{(j)}(1 + 50\sqrt{(m_0/m_i^{(j)})})^{-1} & \text{otherwise;} \end{cases} \tag{17}$$

where m_0 is a 'critical mass' and $\sigma^{(j+1)}$ is the new 'effective density' available for the accumulation process. It is easy to show that this model brings to giant planets in the outer part of the system for every realistic density power law and that the top of mass in intermediate regions can be achieved only by means of some particular density distribution (Isaacman and Sagan, 1977).

The characteristic behaviour of masses in the present solar system can be approached

correcting Equation (17) according to the formula

$$\sigma^{(j+1)} = \begin{cases} \sigma^{(j)} & \text{if } m_i^{(j)} < m_0, \\ 51\sigma^{(j)}(1 + 50\sqrt{(m_0/m_i^{(j)})})^{-1}(1 - \alpha_{pi}^{(j)q}(1 - \sqrt{(m_0/m_i^{(j)})})) & \text{otherwise, (18)} \end{cases}$$

where $\alpha_{pi}^{(j)} = r_i^{(j)}/R$ and q is a free parameter. This last is evidently an arbitrary assumption even if it physically corresponds to the hypothesis of some mass depletion in the outer regions of the system.

5. Results and Discussion

By means of Equation (18) we performed several computations. The parameters of the model are: α_1 , χ , θ , β , μ , m_0 and q . α_1 defines the inner boundary of the planetary system. A too low α_1 value should bring too many small bodies in the proximity of the Sun. To discard this unrealistic situation we assumed an α_1 value of 0.03 corresponding to an inner boundary of the system in the region of terrestrial planets. For χ we choose values in the range 0.8–1.9 corresponding the former to a high mass accretion disk (Cameron, 1978), and the latter nearly to the maximum χ value considered above (and physically corresponding to a purely free-fall disk formation).

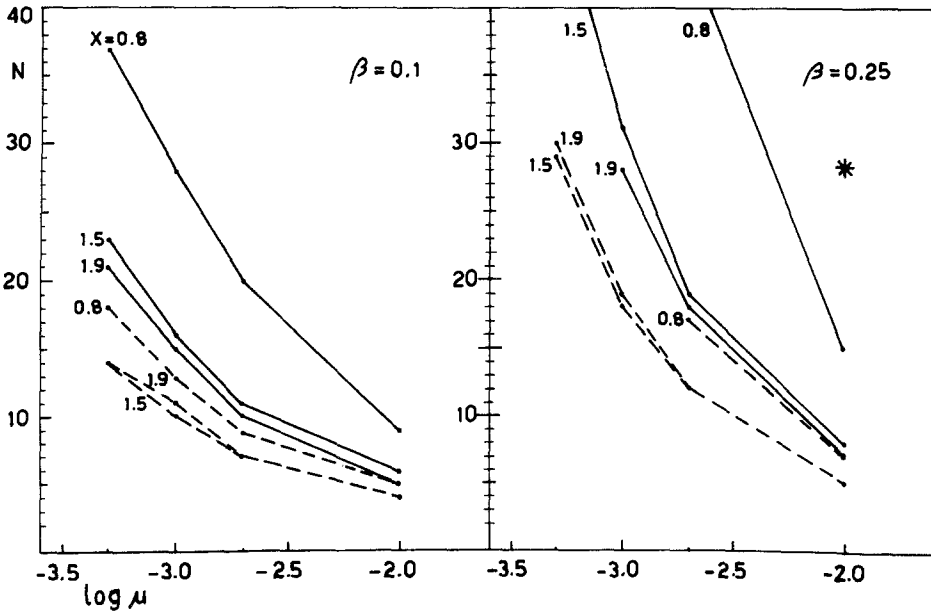


Fig. 1. The figure represents the number of planets (solid line) and that of 'outer' planets (dashed line) as function of $\log \mu$, for various values (labels of the lines) and for two β 's (0.1 and 0.25). The star refers to the case with $\chi = 1.5$, if no gas capture is assumed.

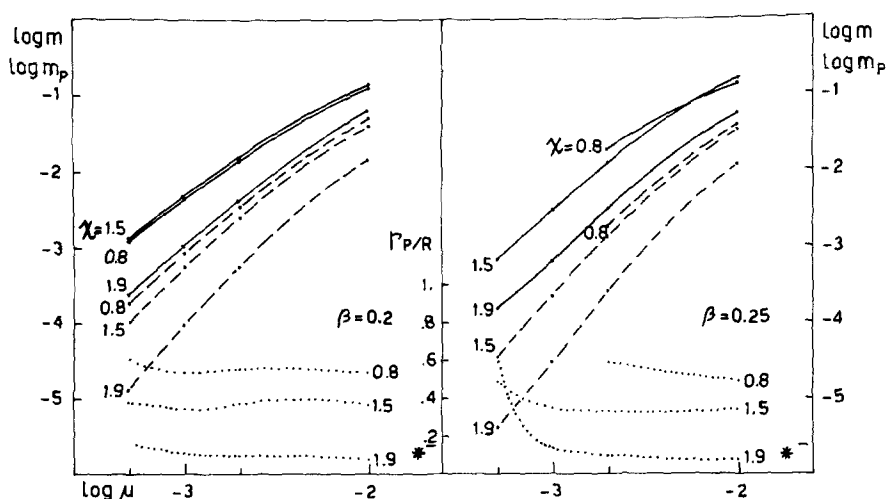


Fig. 2. The figure represents, as functions of $\log \mu$, the total mass of the planets (solid lines) the mass of the major planet (dashed lines) and the semi-major axis of the major planet (dotted lines), for various χ and two β values. The star refers to the present Jupiter distance from the Sun (we assume $R = 40$ AU)

TABLE II

The effect of varying m_0 on output parameters for various cases (symbols are defined in the text)

Model with: $\chi = 0.8$, $\theta = 2$, $\mu = 10^{-3}$, $\beta = 0.25$

m_0	N	N_1	m	m_p	r_p/R
10^{-5}	27	12	0.2451E-02	0.6734E-03	0.62
3×10^{-6}	22	8	0.5128E-02	0.1619E-02	0.455
10^{-6}	17	7	0.9519E-02	0.3228E-03	0.505

Model with: $\chi = 1.5$, $\theta = 2$, $\mu = 10^{-3}$, $\beta = 0.25$

m_0	N	N_1	m	m_p	r_p/R
10^{-5}	13	7	0.2293E-02	0.4041E-03	0.372
3×10^{-6}	9	6	0.6212E-02	0.1334E-02	0.298
10^{-6}	7	5	0.1118E-01	0.2666E-02	0.368

θ and β values are chosen close to Dole (1970) values: θ is 1 or 2, β varies from 0.2 to 0.333 (the power law typical of Hill lobe size).

For μ (the dust mass in solar units) we used four values ranging from 5×10^{-4} to 10^{-2} corresponding to the usual assumptions on the initial nebula mass (Safronov, 1969; Cameron, 1978).

For m_0 we used commonly the value 3×10^{-6} , testing also for lower and higher values; finally we assumed $q = 1$. Figure 1 represents the number of planets (N) and that of 'outer' planets (N_1 : those with $\alpha_p > 0.1$) as functions of parameters χ and μ , for two

TABLE III

The effect of varying θ on output parameters for various cases (symbols are defined in the text)Models with $\mu = 5 \times 10^{-4}$, $\beta = 0.2$, $m_0 = 3 \times 10^{-6}$

x	θ	N	N_1	m	m_p	r_p/R
0.8	1	37	18	0.1197E-02	0.1848E-03	0.612
0.8	2	14	7	0.2116E-02	0.7776E-03	0.492
1.5	1	23	14	0.1158E-02	0.1037E-03	0.383
1.5	2	8	5	0.2278E-02	0.5413E-03	0.307
1.9	1	21	14	0.2385E-03	0.1298E-04	0.167
1.9	2	7	5	0.5065E-03	0.8955E-04	0.101

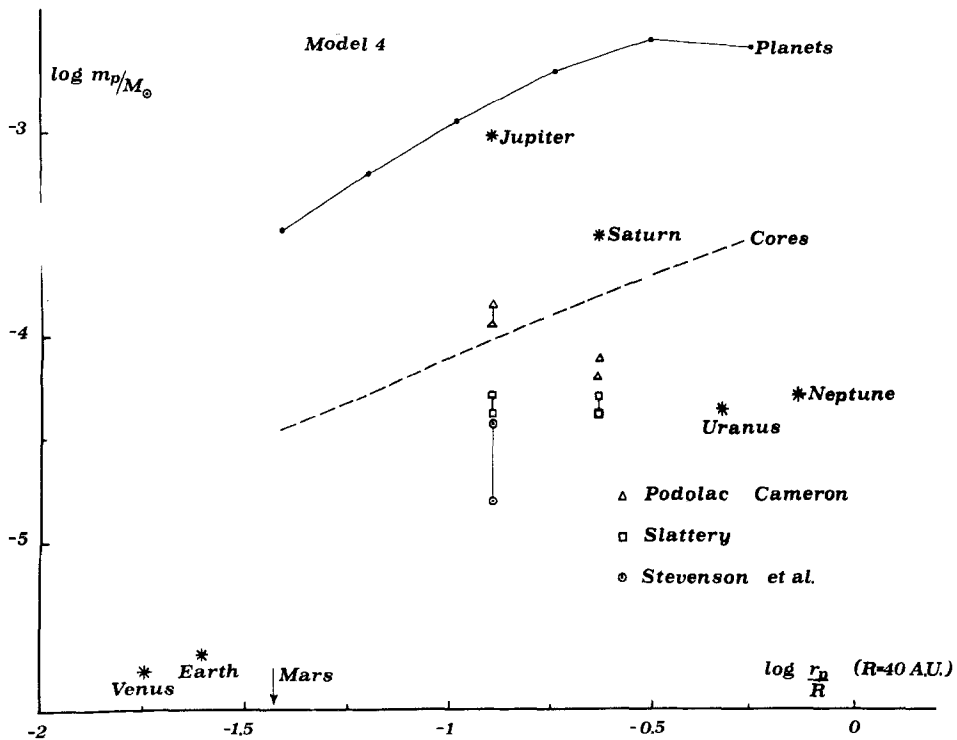


Fig. 3. The figure represents the mass distribution (solid line) and the dust mass distribution (dotted line) for the synthetical model 4, compared with the present solar system and with the results of various authors (Podolak and Cameron (1975), Slattery (1977) and Stevenson *et al.* (1976)).

values of β : the case $\beta = 0.3333$ is not plotted because it gives systems with an unrealistic number of planets (> 60).

Figure 2 represents the total mass (m), the mass (m_p) and the distance from the Sun (r_p) of the largest planet as functions of the same parameters.

Tables II and III select some particular cases to show the effect of varying θ or m_0 .

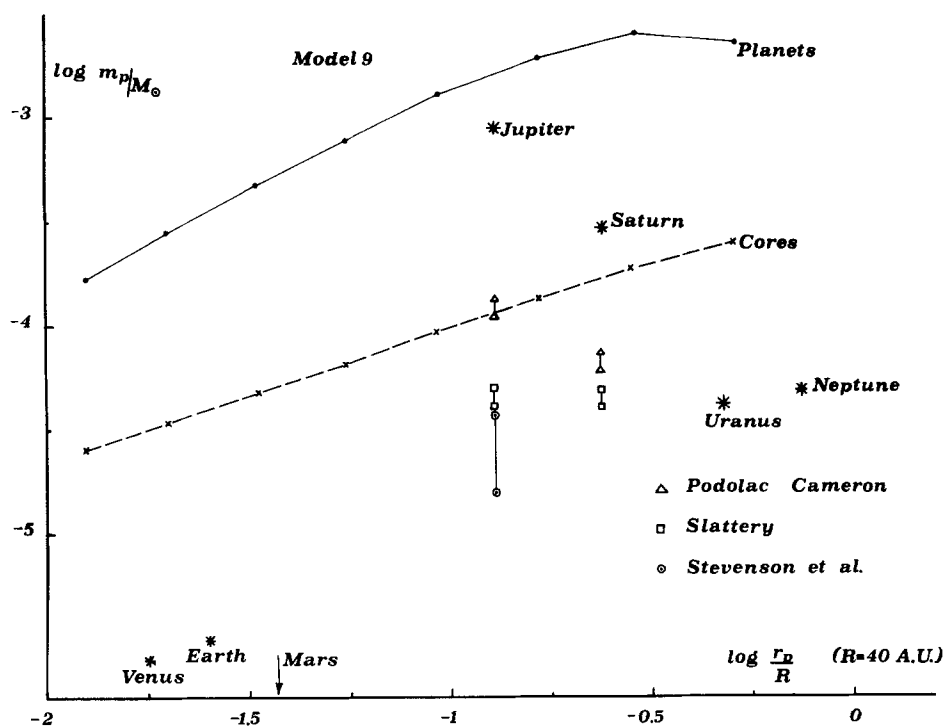


Fig. 4. The same as Figure 3 but with the model 9.

From Figure 1 we can argue that the number of planets and that of 'outer' planets decreases with increasing μ and χ . We remark that the number of planets is in general quite high and attains Solar System-like values only for high χ and μ . Moreover, low β values are strongly supported.

Figure 2 shows that for high μ the percentage of gas captured tends to increase; for low μ and high χ , m is less than μ since we do not take into account the large amount of mass inside the inner boundary of the rings.

In the same figure we show the value of the distance of the largest planet from the Sun: the comparison with the Solar System supports models with a high χ value.

Tables II and III show that the planet number decreases with decreasing m_0 and increasing θ .

6. Best Fit of the Parameters with the Solar System

The large number of free parameters in the model causes a wide range in the characteristics of the output synthetical systems. We attempted to evaluate what choice of the set of parameters would result in the most solar-like system. We used a least-square test for the most meaningful output parameters; they are the total mass of the system, the total mass of the dust, the number of planets and that of the 'outer planets' (in the sense defined

TABLE IVa
The input quantities for the best-fit models (symbols are defined in the text)

Model	α_1	q	χ	θ	μ	β	m_0
1	0.03	0.5	1.8	1.1	1.4E-3	0.139	3.07E-6
2	0.03	1.	1.5	1.1	1.E-3	0.15	3.26E-6
3	0.03	2.	1.8	1.1	1.E-3	0.15	3.53E-6
4	0.03	1.	1.28	1.0	1.1E-3	0.1	2.4E-6
5	0.03	1.	1.28	1.0	1.2E-03	0.125	2.33E-6
6	0.03	1.	1.5	1.1	1.E-3	0.15	3.E-6
7	0.03	1.	1.69	1.1	1.E-3	0.15	3.E-6
8	0.03	1.	1.5	1.1	1.E-3	0.125	3.E-6
9	0.01	1.	1.43	1.0	1.2E-3	0.1	2.85E-6
10	0.05	1.	1.5	1.0	1.4E-3	0.125	2.7E-6

TABLE IVb
Output quantities for best-fit models (lower values are the ones to fit, derived from the present solar system under various assumptions)

Model	m_{tot}	m_{core}	N_{pl}	N_{extpl}	m_{max}	R_{max}	r.m.s.
1	4.15E-3	6.67E-4	7	5	7.72E-4	0.136	0.52
	1. E-3	1. E-5	8	4	9. E-4	0.12	
2	6.05E-3	7.12E-4	8	5	1.22E-3	0.334	0.60
	1. E-3	1. E-5	8	4	9. E-4	0.12	
3	4.32E-3	4.46E-4	7	5	8.02E-4	0.295	0.57
	1. E-3	1. E-5	8	4	9. E-4	0.12	
4	9.5 E-3	7.85E-4	6	4	2.83E-3	0.41	0.39
	1. E-2	1. E-4	8	4	3. E-3	0.36	
5	9.58E-3	8.06E-4	7	4	2.7 E-3	0.41	0.38
	1. E-2	1. E-4	9	4	3. E-3	0.36	
6	6.3 E-3	7.21E-4	8	5	1.27E-3	0.34	0.46
	3. E-3	3. E-5	9	4	1.5 E-3	0.36	
7	5.26E-3	6.45E-4	8	6	9.32E-4	0.27	0.58
	1. E-3	1. E-5	9	4	1. E-3	0.12	
8	7.23E-3	8.17E-4	7	5	1.72E-3	0.37	0.43
	5. E-3	5. E-5	9	4	2. E-3	0.36	
9	1.04E-2	8.72E-4	8	4	2.75E-3	0.38	0.37
	1. E-2	1. E-4	9	4	3. E-3	0.36	
10	1.15E-2	1.05E-3	6	5	2.81E-3	0.37	0.43
	1. E-2	1. E-4	9	4	3. E-3	0.36	

before), the mass and the semi-major axis of the largest planet. The fit varies the values of χ , θ , β , μ and m_0 with selected values of α_1 and q .

Figures 3 and 4 represent some best-fit planetary systems. Table IV compares the out-

put parameters of some best-fit systems with the corresponding ones of the Solar System, showing also the value of the input parameters.

Analysis of input parameters shows that:

(1) The mass is not so high as one could have inferred by the discussion of Section 5 and seems to support a moderate mass depletion, even if more effective than in our model.

(2) High χ values are supported, in agreement with results of Figure 3 about the location of the largest planet; this fact confirms the role of mass depletion in the external region.

(3) β values are too low to be consistent with any reasonable capture model in circular rings; this suggests that eccentricity can play an important role in broadening the capture rings (Vityazev *et al.*, 1978).

7. Conclusions

We can conclude that a general description of the fundamental features of the Solar System can be derived only by very simple assumptions on the matter-capture law.

Moreover, in the one component model it is possible to write an explicit analytical relation between the planets number and the mass of the disk. Such relation remains qualitatively valid also in the gas capture model.

Finally our results confirm the suggestion of a moderate mass depletion in the outer regions of the system as already pointed out by some models of protoplanetary accumulation (Safronov, 1969) and by the 'phase transition model' (Farinella and Paolicchi, 1978; Farinella *et al.*, 1979).

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